How to unlock US$1.3 trillion in resilience finance

Using risk assessment to drive investment and impact

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Climate change is intensifying economic losses worldwide, with the greatest impacts in countries least able to manage them. Yet despite evidence that resilience investments deliver economic and fiscal returns, finance for resilience remains far below what is needed. This paper shows how resilience can be transformed from a perceived cost into an investable asset. The insurance sector's risk analytics and engineering expertise can quantify avoided losses, identify resilience-building opportunities and demonstrate financial returns. Integrating these insights into sovereign finance, multilateral development bank operations and private investment can turn resilience from an undervalued priority into a mainstream, finance-ready asset that is investable at scale.

Contents

Abbreviations

AEL	Annualised expected loss
BCI	Better Cotton Initiative
CAT models	Catastrophe models
COP	Conference of the Parties to the United Nations Framework Convention on Climate Change
DR	Damage ratio
DRFI	Disaster Risk Financing and Insurance
DRM	Disaster risk management
EBRD	European Bank for Reconstruction and Development
EIB	European Investment Bank
EMDE	Emerging markets and developing economies
ESG	Environmental, social and governance
FRLD	Fund for responding to Loss and Damage
G20	Group of 20
GCF	Green Climate Fund
GDP	Gross domestic product
GEFF	Green Economy Financing Facility
IDF	Insurance Development Forum
IDFC	International Development Finance Club
IMF	International Monetary Fund
IoT	Internet of things
IPCC	Intergovernmental Panel on Climate Change
LCOE	Levelised cost of electricity
LDCs	Least developed countries
LEP	Loss exceedance probability
MDBs	Multilateral development banks
OpenDRI	Open Data for Resilience Initiative
PML	Probable maximum loss
ROI	Return on investment
SCM	Synthetic control method
SEADRIF	Southeast Asia Disaster Risk Insurance Facility
SIDS	Small Island Developing States
UNEP	United Nations Environment Programme

Summary

Why essential resilience investment remains constrained

Climate change is driving a steady rise in the frequency, intensity and cost of disasters. In 2024, global disaster losses reached US$320 billion, of which only US$140 billion were insured.[1] Over the past five decades, the number of recorded disasters has increased fivefold and weather-, climate- and water-related events have caused US$4.3 trillion in losses.[2] These impacts are global but fall disproportionately on least developed countries (LDCs) and Small Island Developing States (SIDS), where fiscal capacity is weakest and recovery costs are highest.

Beyond physical damage, disasters create deep economic consequences. They reduce government revenues while driving up spending needs for relief and reconstruction, forcing countries into debt and fiscal stress. Our analysis shows a strong correlation between climate exposure and sovereign debt distress, with the most climate-vulnerable countries facing the highest default-to-debt ratios. Currency depreciation compounds this effect: in 2022 alone, SIDS lost about US$2.4 billion annually and LDCs around US$11.6 billion to exchange rate pressures linked to climate impacts.[3] These recurrent shocks feed directly into credit ratings and borrowing costs, leaving little fiscal space for resilience investment.

The scale of infrastructure at risk and the cost of misinvestment

At the same time, the world faces an unprecedented wave of new infrastructure development. Between 2016 and 2040, the Group of 20's (G20) Global Infrastructure Outlook projects US$94 trillion will be required for investment, with US$15 trillion of this still unfunded.[4] In Africa, nearly 80% of the 2050 building stock has yet to be built;[5] in Asia, the figure is more than half.[6] This represents both a historic opportunity and a serious risk.

If these investments are designed based on historical climate data rather than forward-looking risk scenarios, countries risk locking in trillions of dollars of non-resilient infrastructure that may fail long before the end of its design life. Such misinvestment can turn assets into liabilities, creating systemic vulnerabilities across energy, transport and water networks. The cost of under-preparedness will only multiply with increasing climate risks, whereas designing infrastructure for future risks can protect economies and preserve fiscal stability.

Turning risk modelling into a driver of resilience finance

To break this cycle of underinvestment and avoid locking in fragility, resilience must be viewed as a value-generating investment. Countries that invest early in disaster risk reduction and resilient infrastructure recover faster and sustain stronger credit performance after shocks. Our modelling across eight developing economies found that a 1-in-20-year climate event could lead to US$21.4 billion in losses in total across the eight countries. To cover these losses reactively would cost around US$93 billion, while anticipatory resilience investments could deliver protection against these losses for only US$4.1 billion, nearly 80% less. Every US$1 spent on resilience yields more than US$5 in avoided losses and wider economic benefits.

Demonstrating the value of insurance risk analytics

Despite this evidence, resilience mostly continues to be treated as a cost rather than an investment. Most governments, multilateral development banks (MDBs) and investors still lack credible tools to assess the positive financial return of risk-reduction measures. Conventional assessments focus narrowly on direct damages and short-term recovery costs, while existing climate risk tools are often fragmented or fail to capture the probability and scale of future extremes. Engineering assessments identify physical vulnerabilities but not the broader fiscal or credit impacts, while financial models rarely value the benefits of prevention.

The insurance sector's probabilistic catastrophe (CAT) models and on-site risk engineering can address these gaps. By integrating hazard probabilities, asset exposure and vulnerability data, they simulate thousands of potential events and quantify how specific resilience measures, such as flood defences or stronger building codes, can shift loss curves and generate measurable avoided losses. These models provide granular, forward-looking and monetised insights that can be applied directly to financial decisions.

Our analysis of five representative power plants across Africa, Asia, the Caribbean, the Pacific and Latin America shows that annualised expected losses

can fall from US$48.6 million to US$9.8 million with resilience measures, saving about US$39 million each year. Over 20 years, this amounts to roughly US$10 billion in avoided losses, while extreme event losses would decline by about US$177 million per year. Scaled to the US$8.9 trillion global power sector, resilience investments could prevent US$23 billion in cumulative losses over two decades, a four-to-six-fold return delivered through avoided damage and sustained operation.

Real-world examples underscore this potential. AUDI's flood resilience measures in Germany, developed with Zurich Resilience Solutions, kept production running during the 2021 floods, avoiding millions in losses,[7] while Chennai's 2015 floods, where such safeguards were absent, caused weeks of shutdowns and major economic disruption.[8] Likewise, Madrid's climate-resilient urban design, informed by risk analytics, protected public health and infrastructure,[9] whereas Paris, which lacked comparable planning, saw 735 heat-related deaths during the 2003 heatwave.[10] These cases demonstrate that if the insurance sector offers risk analytics as a standalone service, it can help governments, MDBs and the private sector in viewing resilience as an investable opportunity.

Embedding risk analytics into finance to make resilience investable

Delivering this potential transformation requires coordinated action across governments, MDBs, insurers and investors, each aligning their mandates so that risk analytics inform every stage of financial planning and delivery. Governments can use quantified avoided losses to strengthen sovereign and municipal bond frameworks, issuing green, social or sustainability-linked bonds on stronger terms. MDBs can adopt probabilistic modelling in project appraisal and guarantee design to expand pipelines of bankable projects. Private investors can integrate resilience metrics into environmental, social and governance (ESG) frameworks and due diligence to safeguard long-term portfolio value.

Building such an ecosystem will, however, require accessible data, modelling platforms, regional hubs and continuous capacity-building, particularly for climate-vulnerable countries like LDCs and SIDS. Climate funds and philanthropic partners can help offset the initial cost of supporting access to such analytics and training, ensuring that all countries can apply these tools to guide investment.

Making the resilience gap investable

The Baku to Belém Roadmap to 1.3T, launched under the United Nations Framework Convention on Climate Change Conference of the Parties (COP) presidencies of Azerbaijan and Brazil, provides the political framework to mobilise US$1.3 trillion annually by 2035 for adaptation and resilience.[11] To achieve this ambition, resilience must become a mainstream, finance-ready asset class, one that is embedded across the architecture of global finance: how sovereign bonds are structured, how MDBs appraise and price loans, and how investors evaluate portfolios.

Governments should leverage risk analytics to build credibility with markets and rating agencies, MDBs should integrate probabilistic risk modelling across their operations and private investors should reward projects that demonstrate quantifiable risk-reduction measures. If Baku marked the political commitment, Belém must set the operational standard, establishing risk analytics as the foundation for scaling finance.

By integrating these approaches across sovereign, multilateral and private investment systems, resilience can move from being underfunded and undervalued to being investable at scale, closing the resilience gap not through one mechanism but by ensuring that risk-informed decision making becomes central to how global finance operates.

Why essential resilience investment is not reaching the level needed

Climate change impacts are increasing — and so are losses, particularly for SIDS and LDCs, where resilience and coping systems are weakest. This section examines how climate disasters erode the fiscal stability of vulnerable countries by creating spikes in expenditure, deepening reliance on debt and widening fiscal deficits. Investing in resilience can provide vital protection and prevent small shocks from spiralling into crises, but our research shows that disaster-induced macroeconomic pressures are pushing countries into a vicious cycle of underinvestment in resilience. Meanwhile, private and international public finance are disincentivised from investing and funding remains far below what is needed.

1

1.1 Climate change impacts are increasing — and so are losses

Climate change is driving an increase in the frequency, intensity and scale of natural catastrophes. 2024 was the warmest year in the 175-year record, with global average temperatures 1.55°C above pre-industrial levels for the first time, as well as record ocean heat and rising sea levels that amplified extremes across regions.[12] The Intergovernmental Panel on Climate Change (IPCC) has confirmed with high confidence that even a small amount of additional warming significantly increases hot extremes, heavy rainfall and drought intensity,[13] extending risks into areas previously considered low hazard.[14]

The impacts are evident in loss figures. Global disaster losses in 2024 reached US$320 billion, of which about US$140 billion were insured, making it one of the costliest years on record.[15] Over the past five decades, the economic cost of weather, climate and water-related disasters has reached US$4.3 trillion, with the number of recorded disasters rising fivefold since 1970.[16] The annual average cost of disasters almost doubled when comparing the periods 1980–1999 and 2000–2019. This increase is largely driven by climate-related disasters, which account for over 90% of all major recorded disasters.[17]

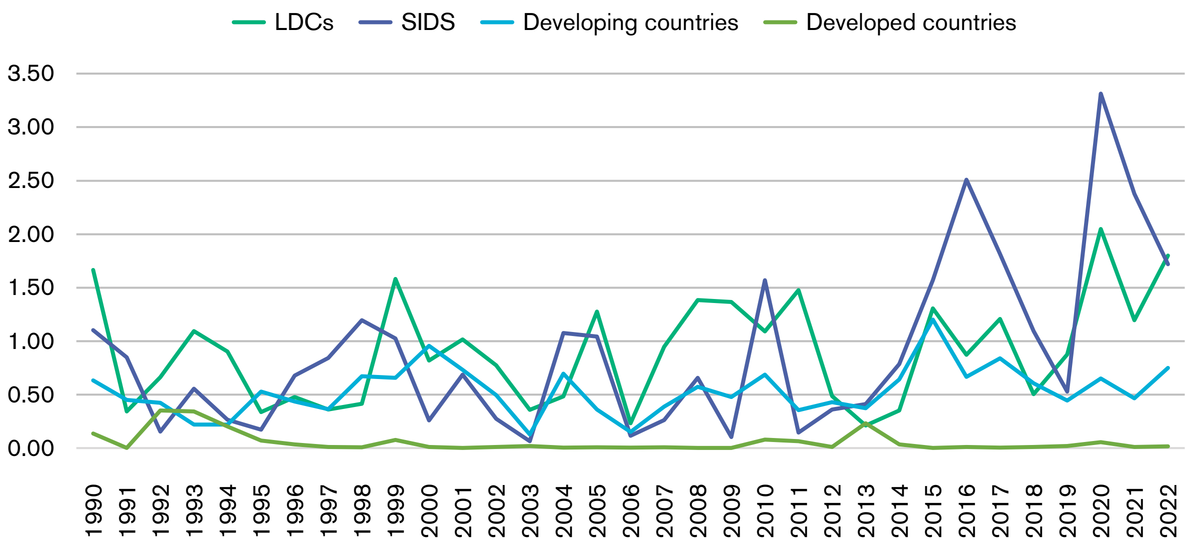
In Figure 1, we show how the disaster intensity trend, while increasing universally, has increased unevenly across different country groups, with SIDS and LDCs most impacted. The long-term decadal averages show that the disaster intensity in fragile and conflict-affected states has nearly tripled between 1960–1990 (0.30) and 1991–2022 (0.84). LDCs and SIDS started from a much higher baseline than other countries and have remained consistently exposed, with average disaster intensity in the range of 0.8 and 0.9, respectively, across these periods. In developing countries, disaster intensity almost doubled (0.28 to 0.53), while in developed countries, intensity remained low but still tripled (from 0.02 to 0.06). These figures suggest that risks are rising everywhere, but they are most severe where resilience and coping systems are weakest, such as those in LDCs and SIDS.

1.2 Physical losses due to climate impacts are leading to economic consequences

The rising intensity of climate disasters not only translates into increased physical damage. It also creates profound economic consequences that undermine the ability of climate-vulnerable countries like LDCs and SIDS to invest in development and resilience.

One of the most direct channels through which climate change affects economies is via its impact on debt. Disasters reduce government revenues while creating sudden spikes in expenditure for relief and reconstruction. Countries often rely on additional borrowing to bridge this gap, and repeated shocks force governments to layer debt on debt, creating a cycle of rising indebtedness and reduced fiscal space. Our analysis shows a strong relationship between climate exposure and debt distress: analysis of 71 countries demonstrates that those with a higher hazard and

Figure 1. Disaster intensity in SIDS, LDCs, and developing and developed countries



Source: Author assessment based on data from EM-DAT (EM-DAT, n.d.)[115]

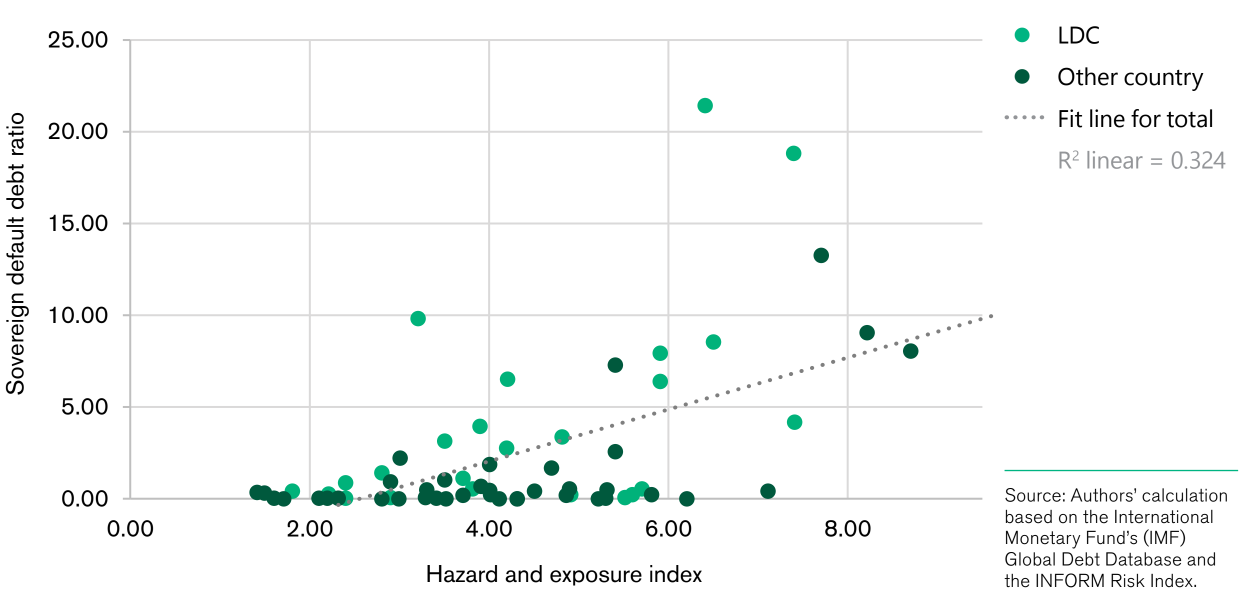
exposure index also have a higher sovereign default-to-debt ratio, with the correlation especially strong in LDCs (0.601)[18] (see Figure 2).

Beyond debt, disasters disrupt growth, widen fiscal deficits and destabilise revenues. Our analysis shows that, in SIDS, fiscal deficits widened from an average of –2.8% of GDP during low disaster years to –4.5% of gross domestic product (GDP) during high-intensity years, while tax revenue volatility reached 17.5%.[19]

Currency depreciation due to climate impacts compounds these pressures by raising the cost of servicing debt denominated in foreign currency and inflating import prices. Our regression analysis shows that for every one unit increase in climate vulnerability, the exchange rate of countries against the US$ rises by 65%.[i] The economic costs of this are substantial. In 2022 alone, the average annual exchange-related loss per country was US$2.4 billion for SIDS and US$11.6 billion for LDCs, while cumulative losses between 1991 and 2022 reached US$27 billion for SIDS and US$68 billion for LDCs[20] (see Figure 3). This is a substantial amount that could have been used for resilience building.

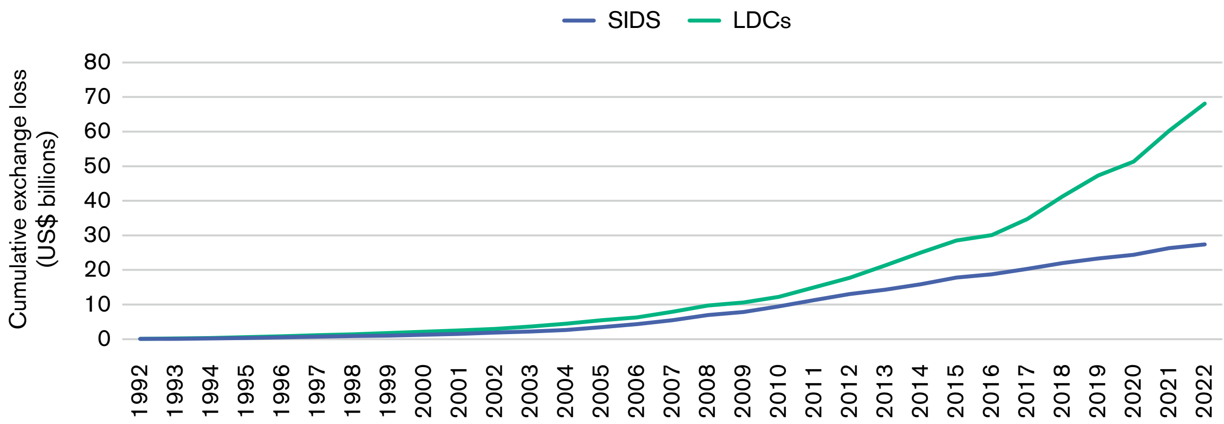
As disasters weaken growth, worsen fiscal balances and increase debt burdens (all indicators that rating agencies use to assess a country's creditworthiness), these macroeconomic stresses feed directly into how

Figure 2. Relationship between hazard and exposure index and sovereign debt default



Source: Authors' calculation based on the International Monetary Fund's (IMF) Global Debt Database and the INFORM Risk Index.

Figure 3. Cumulative exchange-related losses for SIDS and LDCs

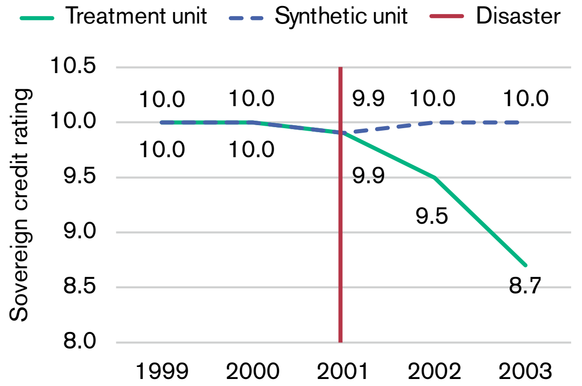


Source: Calculation based on IMF International Financial Statistics (IFS).

i More details on regression analysis can be found in Bharadwaj, R, Karthikeyan, N and Kumar, BA (2025) Currencies under pressure: how currency fluctuations and climate risks impact debt sustainability in SIDS and LDCs. IIED, London. www.iied.org/22626iied markets and rating agencies assess vulnerability. We used the synthetic control method (SCM)[ii] to understand this effect clearly. In Grenada, Hurricane Ivan in 2004 caused damage equivalent to 200% of GDP. While the actual credit ratings dropped from 8.2 to 1.6 in a single year, our assessment showed that the synthetic counterfactual (without the disaster) remained at 7.8[21] (see Figure 5).

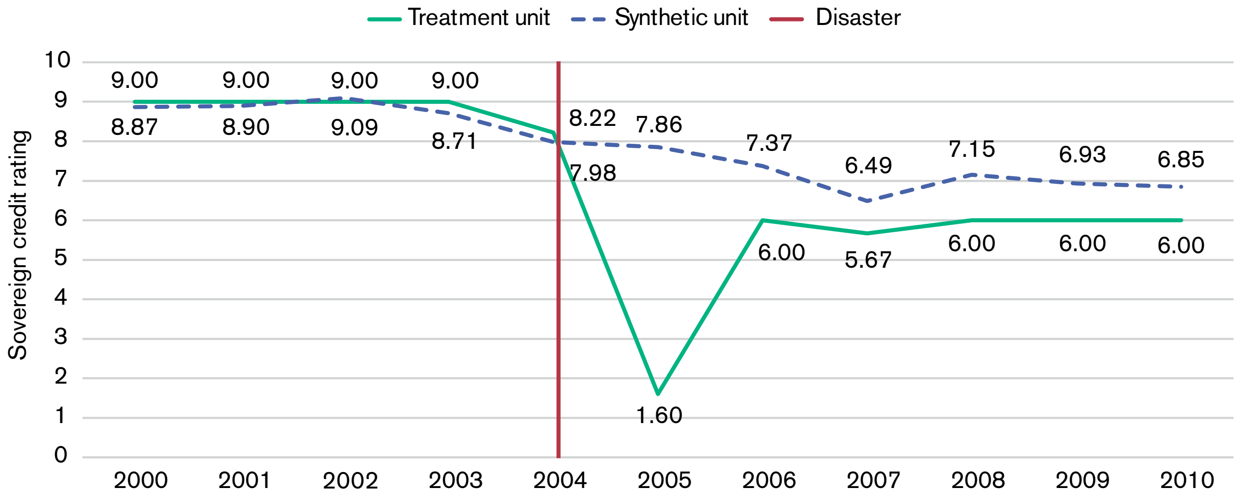
We found similar patterns in Belize and Papua New Guinea, where ratings fell sharply after disasters. Had these countries not been hit by disasters, their credit ratings would have remained more stable, as demonstrated by their synthetic counterfactual without disaster (see figures 4 and 6).

Figure 4. SCM analysis of Belize's sovereign credit rating



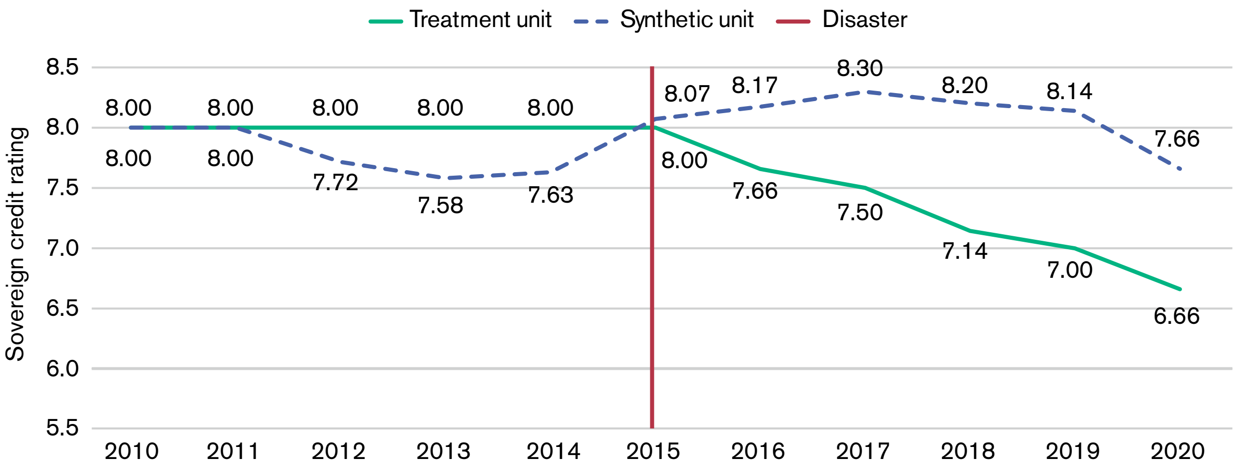
Source: Authors' calculation based on data from Kose et al. 2022.[116]

Figure 5. SCM analysis of Grenada's sovereign credit rating



Source: Authors' calculation based on data from Kose et al. 2022.

Figure 6. SCM analysis of Papua New Guinea's sovereign credit rating



Source: Authors' calculation based on data from Kose et al. 2022.

ii The SCM is a statistical technique used to estimate what would have happened without a specific event, such as a climate-related disaster. It does this by creating a 'synthetic' version of the affected country, built from a weighted mix of other similar countries and then comparing outcomes to isolate the disaster's true impact.

1.3 The funding challenge: why resilience remains under-financed

Investing in resilience by constructing stronger housing, roads, bridges, flood defences, drought-proofing and taking ecosystem-based measures can lower hazard exposure, stop small shocks from spiralling into crises and provide a buffer against macroeconomic shocks. However, disaster-induced macroeconomic pressures push countries into a vicious cycle of underinvestment in resilience (see Figure 7).

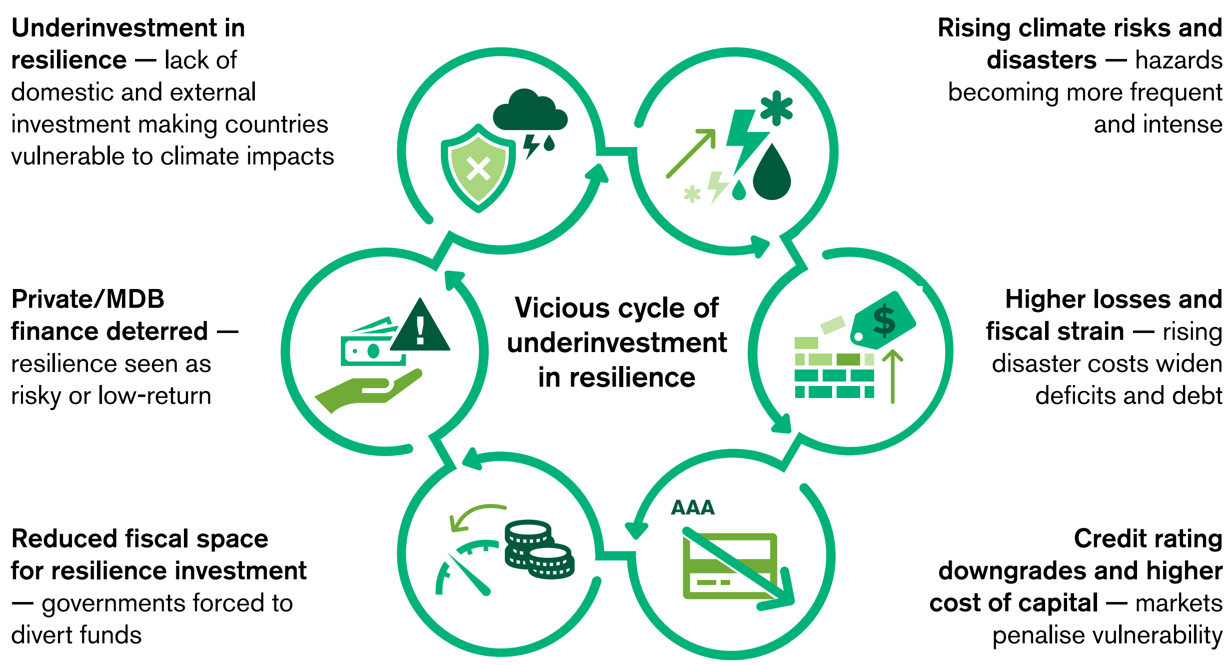
Climate risk added more than US$40 billion to interest payments for the 40 most exposed nations between 2007 and 2016 and could cost US$168 billion more over the next decade (2017–2026).[22] By 2030, as many as 63 sovereigns may face climate-driven credit rating downgrades, adding between US$22 billion and US$33 billion annually to debt service costs.[23] For governments already struggling to maintain essential services, the scope for prevention and resilience investment becomes severely constrained due to these impacts.

International climate and development finance has failed to fill the gap. Adaptation needs in developing countries are estimated at US$215–387 billion per year this decade, but tracked flows were only US$21–63 billion in 2021–2022.[24] Climate change loss and damage needs are projected to be US$290–580 billion annually by 2030, while pledges to the Fund for responding to Loss and Damage (FRLD) remain in the millions.[25] Humanitarian finance is similarly strained, with UN appeals funded at only about 40% in recent years.[26] Broader development needs are even larger, with the financing gap for the Sustainable Development Goals now estimated at US$4 trillion annually.[27]

While the private sector and MDBs are acknowledged as essential to bridging the resilience finance gap, their current contributions remain acutely constrained. For instance, although total global climate finance for adaptation reached an annual average of US$63 billion (out of total climate finance of approximately US$1.3 trillion) in 2021–2022, tracked private sector flows into adaptation accounted for less than 3% of this amount.[28] In addition, their investment was heavily skewed toward direct project debt (loans to individual projects), with finance mobilised through capital markets amounting to just US$638 million.[29] MDBs are scaling up, but their contributions are still well below what is required. In 2023, they provided US$74.7 billion in climate finance to low- and middle-income countries, of which US$24.7 billion went to adaptation.[30]

One reason why countries cannot use existing climate finance to leverage private sector participation is that they are not able to demonstrate clear positive returns for resilience projects. Investors therefore perceive them as low-return or high-risk, and private capital

Figure 7. The vicious cycle of underinvestment in resilience



does not often materialise at the required scale. At the same time, climate-driven credit downgrades reinforce negative market signals. This is exacerbated by the fact that current credit rating methodologies largely penalise a country's risk exposure without positively factoring in the protective investments it makes in risk reduction and adaptation.[31] Lower ratings not only increase governments' borrowing costs but also make MDBs and private investors more cautious, narrowing the pool of viable projects. In effect, the very countries that most need private and MDB investment in resilience are those that markets see as least attractive.

A global protection gap also persists alongside resilience investment gaps. In 2023, only 38–40% of US$280 billion in disaster losses were insured, with protection far lower in emerging economies.[32] In 2024, insured losses were US$140 billion, again leaving households and governments to carry most of the cost.[33]

These dynamics together reveal why resilience remains under-financed. The rising costs of disaster recovery and domestic fiscal pressures reduce governments' ability to act, international public finance is far below need, and private and MDB flows are both limited and structurally disincentivised. Climate fund conditionalities, weak investor incentives and the downward pressure of credit downgrades all reinforce a cycle where higher risks suppress investment, underinvestment deepens vulnerability and each new disaster amplifies the fiscal and financial strain.

Unlocking investment: turning risk modelling into a driver of resilience finance

In the previous section, we showed why resilience is both essential and underfinanced. In this section, we shift the focus from the constraints to the opportunities. We first show the risks of underinvestment and misinvestment, where infrastructure designed for yesterday's climate locks in future vulnerabilities. We then show how countries that invest early in disaster risk reduction and resilient infrastructure achieve better outcomes, protecting growth, economic stability and creditworthiness. We then introduce the insurance sector's risk assessment expertise as a mechanism to turn resilience into an investable proposition by quantifying avoided losses, outlining resilience dividends and enhancing return on investment, providing the kind of evidence that governments, MDBs and private investors typically need to unlock resilience finance at scale.

2

2.1 The double risk of underinvestment and misinvestment

Much of the infrastructure on which developing and emerging economies will rely in the coming decades has not yet been built. This creates a unique opportunity to ensure that new investments are designed to be resilient and future-fit, rather than locking in assets that are fragile in the face of future climate risks. The G20's Global Infrastructure Outlook estimates that US$94 trillion in infrastructure investment will be needed globally between 2016 and 2040, with US$15 trillion of this still unfunded under current trends.[34] These investments span water and sanitation, electricity, transport, and telecommunications. In developing countries, the scale of construction still needed is even more striking. The United Nations Environment Programme (UNEP) projects that around half of the building stock required globally by 2050 has yet to be built,[35] while in Africa the figure is closer to 80%.[36] Whether this infrastructure pipeline is designed to be resilient and future-fit, or instead locks in fragility, will to a large extent determine the economic and fiscal stability of these countries for decades to come.

Yet in practice resilience is often misunderstood and underfunded. As a result, two damaging patterns can emerge. First, countries risk locking in fragile, non-resilient infrastructure that will not be able to withstand future weather extremes, creating assets that become liabilities well before the end of their design life. Second, even where resilience measures are included in the infrastructure design, they are often based on historical averages rather than forward-looking risk assessments. These designs typically assume the climate is static and weather patterns remain unchanged. However, in a warming world, we are seeing more frequent and intense storms, sea-level rise and compound events.[37] More importantly, climate change is shifting the probability distribution: previously rare and catastrophic outcomes (often called 'tail risks') are becoming more common, making current investment in resilience a necessity. Without probabilistic assessments that account for a range of possible futures, including extreme but increasingly frequent events, countries risk investing in infrastructure which appears robust on paper but which may fail in reality.

The costs of under-preparedness are already evident, particularly where ageing infrastructure, designed without accounting for evolving climate risks, has led to cascading failures. In the northern New Jersey estuarine system, for instance, coastal berms and tide gates built decades ago based on historical hydrologic data were overtopped during Hurricane Sandy in 2012. This allowed the storm surge to spread 36.2km inland along the Hackensack River and threatened the Oradell Dam, constructed in 1923.[38] This compound event of surge and precipitation exposed vulnerabilities in interconnected water, transport and energy systems, disrupting services for millions of people in one of the US's largest metropolitan areas and highlighting how short-term hydrologic records failed to capture intensified rainfall and surge under non-static conditions.[39] Similarly, Hurricane Helene in 2024, driven by Gulf of Mexico sea surface temperatures that were up to 2°C (3.6°F) above average, caused at least 10% heavier rainfall due to climate change, triggering unprecedented flash flooding that devastated inland infrastructure in the Southern Appalachians, including roads, bridges and power grids, leaving 4.5 million people without power and causing at least 250 deaths.[40,41] These failures underscore how infrastructure predicated on past climate data succumbs to high-intensity disasters, amplifying disruptions and prolonging recovery.

Looking ahead, these costs will only rise. Ratings agency S&P Global projects that corporate physical risk costs could reach US$1.2 trillion annually by the 2050s, primarily from climate-driven disruption to operations, supply chains and assets.[42] In the US, cumulative losses from weather and climate-related disasters since 1980 already exceed US$2.9 trillion, while the European Union has faced steadily rising losses over the same period, totalling over €738 billion through 2023.[43,44] These figures underscore that embedding resilience in infrastructure planning now can prevent assets from becoming stranded, protect economies from escalating shocks and unlock higher long-term returns.

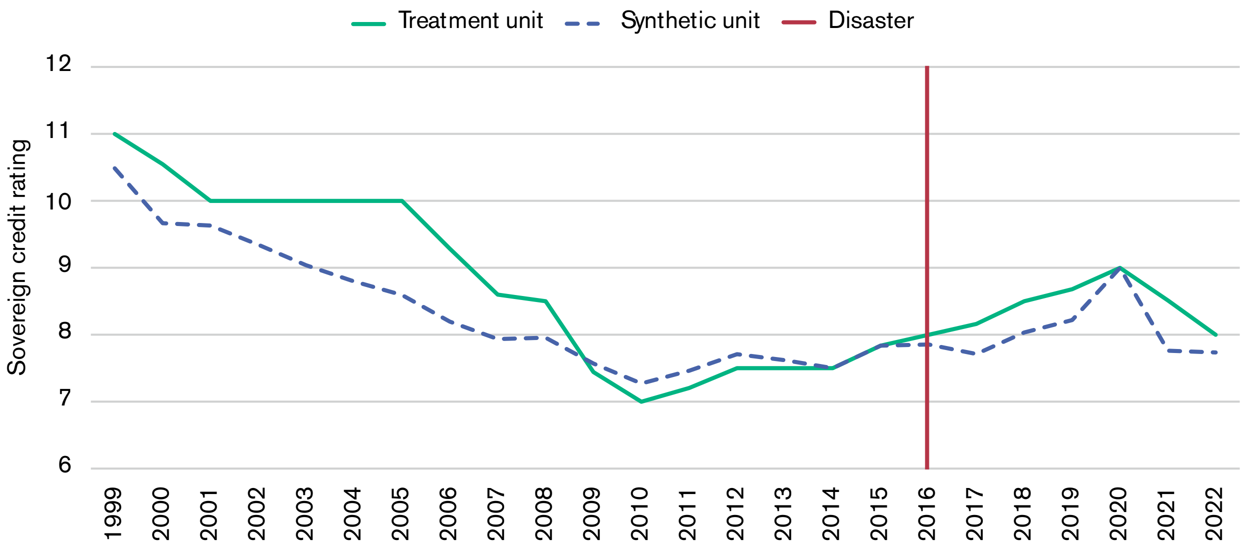
2.2 Countries that invest in resilience are better prepared

Evidence shows that countries that invest in disaster risk management and resilient infrastructure earlier are better able to withstand shocks and recover more quickly. They not only reduce losses but also sustain growth, protecting fiscal space and strengthening investor confidence.

One clear example of this is Fiji. Despite being highly exposed to some of the world's most intense cyclones, the country has invested strategically in preparedness and resilience. When Cyclone Winston struck in 2016, damages reached approximately US$900 million (about 20% of GDP) and nearly 60% of the population was affected. Yet Fiji stabilised its fiscal balance, avoided sovereign default and maintained a stronger credit rating. Some of these proactive measures included investments in climate-resilient roads, water infrastructure upgrades, improvements in early warning systems and the establishment of a dedicated emergency contingency fund to ensure a rapid financial response after a disaster.[45] SCM analysis presented in Figure 8 confirms this. Fiji's ratings remained similar or above its synthetic counterpart from 2016 onwards, in sharp contrast to prolonged downgrades suffered by other SIDS such as Belize, Grenada and Papua New Guinea after comparable disasters (see figures 4, 5 and 6). This demonstrates that resilience investments can buffer long-term financial deterioration by protecting fiscal stability and market confidence.

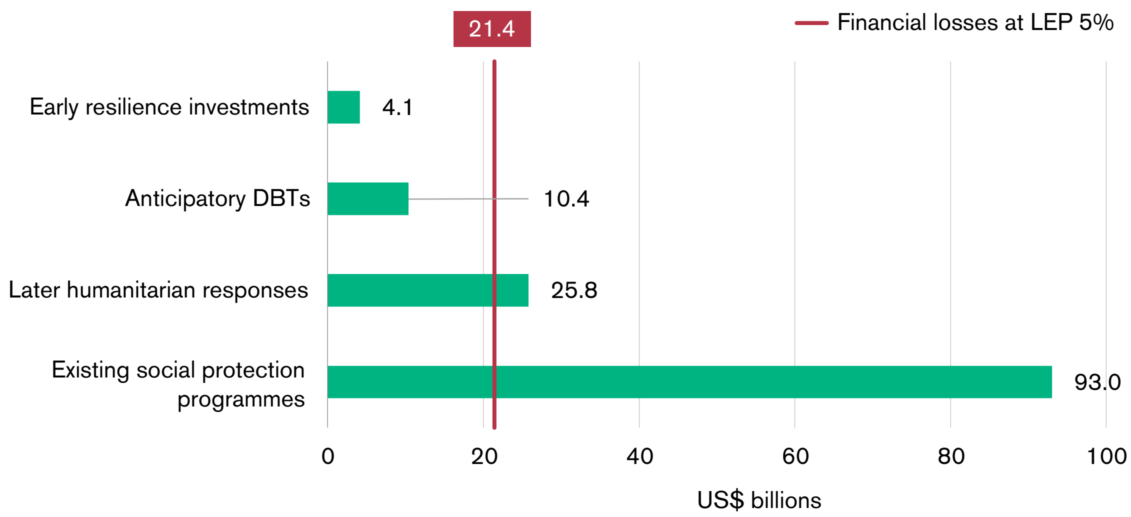
The broader economic case becomes clearer when we compare the costs of resilience with the costs of inaction. Using loss exceedance probability (LEP)[iii] analysis across eight developing economies selected to represent a diverse range of climate risk contexts[46] (Bangladesh, Ethiopia, Ghana, India, Malawi, Pakistan, Senegal and Uganda), we estimate that a 1-in-20-year event (5% LEP) would generate losses of US$21.4 billion. Covering these losses through existing humanitarian or social protection systems would cost nearly US$93 billion, more than four times the losses themselves. By contrast,

Figure 8. SCM analysis of Fiji's sovereign credit rating



Source: Authors' calculation based on data from Kose et al. 2022.

Figure 9. Cost of covering disaster-related losses through existing systems versus early resilience investment across eight countries



Source: Authors' calculations.

iii LEP estimates the probability that financial losses will exceed a given threshold in any year. A 5% LEP reflects a 1-in-20-year event (severe but plausible), while a 50% LEP represents more frequent, lower intensity shocks.

early resilience investments could deliver the same level of protection for just US$4.1 billion, nearly 80% less than the cost of a reactive response (see Figure 9). The cost–benefit ratio of resilience-building interventions averaged 5.17, meaning that every US$1 invested yields more than US$5 in avoided losses and wider economic benefits.[47]

These findings are consistent with broader global evidence on the high returns of resilience investments. A recent JP Morgan analysis synthesises studies showing that every US$1 spent on climate adaptation and resilience yields an average return of between US$2 and US$43 in economic benefits. The analysis covers a mix of economy-wide and corporate sector benefits, with specific examples including US$4 from resilient infrastructure, up to US$43 in the food and beverage sector and US$13 from comprehensive preparedness efforts. These returns often materialise within 12–24 years, complementing longer-term mitigation gains and transforming adaptation into a catalyst for innovation and market positioning.[48]

Resilience also generates wider dividends that go beyond avoided losses. Our modelling suggests that if SIDS invested systematically in resilience (for example, through upfront measures like resilient infrastructure) and insurance protection measures (for example, in risk transfer products such as parametric insurance for post-disaster liquidity), their average credit rating could improve by nearly one point, reducing borrowing costs and improving access to finance.[49] This finding is consistent with a growing body of research and data initiatives focused on better reflecting adaptation efforts in sovereign financial metrics.[50] As shown by our assessment, GDP growth, which averaged −0.79% for the 13 SIDS that currently have a credit rating, could shift into positive territory at +1.17% once resilience investment and insurance coverage are factored in[51] (see Figure 10).

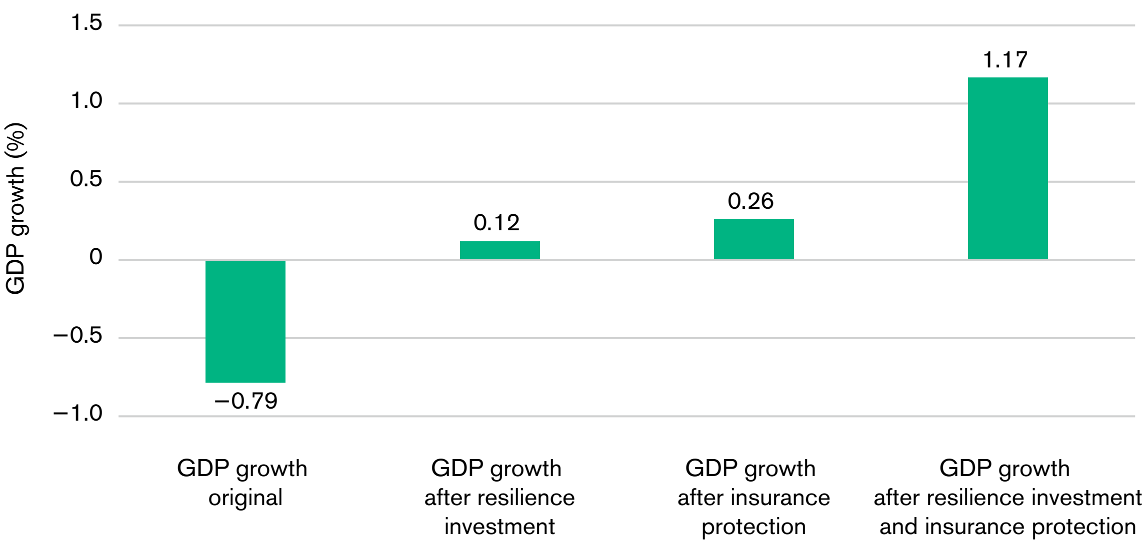
Evidence thus clearly shows that in an era of intensifying shocks, proactive resilience can be the foundation of long-term prosperity, unlocking returns that far outweigh the costs of inaction.

2.3 Unlocking resilience finance through physical risk assessment

Despite evidence that integrating resilience measures into new and existing projects pays off, government, private capital and MDB investment in resilience remains modest.[52] One reason is that the business case for risk reduction and resilience is not clearly demonstrated in financial terms to support investment decision making. Governments often view resilience as a fiscal burden amid competing priorities. Companies, on the other hand, see it as an additional investment cost without a quantifiable monetary return. Similarly, MDBs and banks assess such investments through a broader lens of credit risk, where climate hazards may represent only a small portion of overall concerns, such as political and governance risks. In these contexts, quantification of how specific risk reduction and resilience-building measures can reduce losses is often lacking.

A recent survey of global firms found that 94% now disclose climate-related information, but only 36% reference climate-related financial impacts in their financial statements.[53] As a result, while forward-looking climate risks are acknowledged, existing assessments

Figure 10. Simulated change in GDP growth rate (%) in SIDS after incorporating resilience investment and insurance protection measures



Source: Authors' calculations.

often fail to capture their true materiality, so they are still insufficiently priced into decisions. The positive benefits of risk-reduction investments are largely ignored, or, where measurable, are not consistently factored in, leaving the financial case for resilience unaccounted for. This market failure mirrors the gap in public sector planning, where there is an absence of quantified, forward-looking climate risk assessments and analysis of avoided losses. This reinforces the perception that risk reduction and resilience are additional costs rather than value-generating investments. Even MDBs, despite mandates for climate integration, frequently apply standardised templates that prioritise mitigation over adaptation. Some 69% of the US$85 billion in climate finance allocated to low- and middle-income countries in 2024 was for mitigation.[54] This is often accompanied by limited use of dynamic vulnerability modelling that could show how resilience upgrades can enhance project viability under multiple risk scenarios. World Bank evaluations have also highlighted persistent enabling environment constraints, such as inconsistent risk screening in public–private partnerships — a practice that often leads to public entities bearing disproportionate uncertainty, particularly climate-related risk.[55]

It remains difficult to establish a credible, quantified view of the benefits accrued, losses avoided, economic stability gained and investment returns secured, set against the costs of investing in risk reduction. Without this, resilience will continue to be undervalued and the investment gaps will persist. To drive government, MDB, bank and private sector investments, physical risk assessments must go beyond basic hazard mapping or qualitative evaluations. They need to incorporate probabilistic, multi-hazard approaches that model the full spectrum of potential losses from climate hazards, including tail risks.

Several climate risk assessment tools already exist in the market to support such evaluations, but they fail to capture the full picture. For instance, climate model projections from sources like the World Climate Research Programme or Coupled Model Intercomparison Project (CMIP) offer forward-looking scenarios for hazards such as temperature extremes or precipitation changes. These are often integrated into tools like the World Bank's Climate Risk Analytics or the EU's Copernicus Climate Change Service and they are also used by the IPCC in their assessment reports.[56] Engineering firms provide asset-level vulnerability assessments using international standards like ISO 14091 and also incorporate geospatial data for site-specific exposures. But these standards are often limited as they rely on static risk factors and do not fully capture the dynamic impact of future climate change on vulnerabilities and exposures. Financial institutions and consultancies, such as those using S&P's Trucost or Moody's climate-adjusted credit models, link physical risks to economic indicators like GDP impacts or borrowing costs.[57,58,59] However, these tools often remain siloed and do not take into account the positive impact of investing in risk reduction. Climate models may average uncertainties without capturing tail risks; engineering assessments might lack probabilistic depth; and financial models frequently overlook cascading effects on whole economic systems (for example, a flood disrupting power and halting exports), compound hazards (for example, droughts and heatwaves), socioeconomic amplifiers (such as population growth in vulnerable areas) or lack of granular risk-reduction interventions, leading to underestimated systemic vulnerabilities and outputs that are either too generalised or not fully actionable for investment decisions.[60,61] Cost–benefit analyses, when applied, typically focus on direct damages from climate hazards while excluding broader benefits from risk-reduction investments, such as reduced fiscal volatility or avoided productivity interruptions.

Advanced risk assessment capabilities, similar to those already available in the insurance sector, offer a pathway to bridge these gaps. They provide granular, forward-looking physical risk assessments that can help highlight the return on investment of risk-reduction measures. The insurance sector is also well-positioned to move quickly on climate risk analytics by leveraging its decades of expertise in probabilistic modelling,[62] advanced data tools like AI-driven simulations and established partnerships to scale applications beyond traditional markets.[63] To date, these tools have been used mainly to inform underwriting decisions. However, there is an opportunity for them to be deployed much more widely as standalone tools, such as through initiatives like the Global Risk Modelling Alliance[64] (a service backed by the V20 Group of Ministers of Finance of the Climate Vulnerable Forum) and the Insurance Development Forum (IDF). Opportunities to use these standalone tools are particularly strong in jurisdictions where insurance markets are less developed and where their use can extend beyond traditional risk transfer to support resilience planning and financing.

2.3.1 Why insurance sector expertise in risk modelling can inform resilience investment

Unlike the deterministic or qualitative methods that dominate most public planning decisions, the insurance sector has developed and refined probabilistic catastrophe (CAT) models and risk engineering tools over several decades (see Box 1). These were originally designed to support underwriting and capital allocation decisions and meet regulatory requirements. However, their analytical depth makes them highly relevant for informing resilience investment more broadly, such as in public infrastructure planning, project financing and corporate risk management.

Many insurers complement probabilistic models with granular site-level risk engineering. This involves inspections of construction quality, building materials, drainage capacity, maintenance practices and overall systemic context to identify vulnerabilities and recommend practical interventions, such as retrofitting bridges for seismic resilience or elevating substations to prevent flooding. The additive value of combining risk modelling with risk engineering is critical. While modelling alone might yield a heat map of risks, integration with risk engineering provides precise, context-specific loss metrics and advice on risk reduction through specific resilience measures, making the outputs more quantifiable, practical and actionable.

In recent times, the insurance sector has developed various tools to expand these assessments beyond core modules, incorporating economic and systemic dimensions that other actors typically don't address. These include:

- Macro-level multi-hazard probabilistic risk assessment, which evaluates risks across entire economies or regions, linking hazards to broader indicators like GDP volatility or fiscal deficits.[65]
- Physical asset-level risk assessment, incorporating multiple parameters such as material strength, location-specific exposures and climate scenarios to project the likelihood and magnitude of losses under different conditions.[66]
- Practical insights on site-specific risk reduction and resilience-building measures and their effectiveness, derived from practical engineering experience and loss projections that compare baseline vulnerabilities with post-intervention outcomes.[67]
- Financial assessments tied to these models, enabling economic evaluations such as cost–benefit ratios or return on investment (ROI) calculations based on avoided losses.[68]
- Comprehensive loss assessments that encompass direct damages, business interruptions and cascading effects, providing a level of detail and understanding of system-level impacts not commonly found in assessments from engineering, climate science or financial sectors alone.[69]

These features give the insurance sector's risk management approach two advantages that make it particularly useful for mobilising resilience finance. First, it is forward-looking: risk reduction and prevention are core pillars of the insurance sector's work, so the sector not only provides reactive coverage but also proactive strategies to minimise losses. Many models incorporate climate change scenarios from the IPCC and other scientific sources, enabling stress tests of infrastructure against future rather than past conditions. A study by Lloyd's, for instance, shows how updated CAT models for US hurricanes now project nearly 50% higher insured losses under a 2°C global warming scenario, informing adaptive strategies.[70] Insurers also act as key aggregators of risk data, compiling vast datasets from claims, exposures and hazards to provide a comprehensive view of what risks look like across regions and sectors, which can inform broader resilience planning.

BOX 1. WHAT DO CAT MODELS AND RISK ENGINEERING COVER?

CAT models are specialised tools used primarily in the insurance and reinsurance industries to assess and quantify the risk of large-scale disasters. They typically integrate three components: hazard data (the probability and intensity of floods, storms, earthquakes or droughts under current and projected climate conditions), exposure data (the location, value and characteristics of assets, such as homes, factories or transport networks) and vulnerability functions (how different asset types perform when exposed to hazards). Running thousands of simulations across these modules generates loss curves, which provide a full distribution of potential losses, from frequent low-level shocks to rare but catastrophic tail risks. The computational power of advanced CAT models now routinely simulates millions of stochastic events per run, drawing on vast datasets from satellites, 'internet of things' (IoT) sensors and historical records to produce robust probabilistic outputs that public tools often lack in scale and granularity. This allows analysts to move beyond average expected losses and quantify the avoided damages that risk-reduction measures (such as flood defences or stronger building codes) can deliver by generating loss curves. These curves enable sensitivity analyses to test how varying resilience interventions shift the entire loss tail, providing a clearer view of cost–benefit trade-offs.

Risk engineering refers to the systematic evaluation and mitigation of risks at the asset or site level, often involving on-site inspections, data collection and recommendations to reduce vulnerabilities. It complements CAT modelling by providing granular, practical insights that refine model inputs and outputs. For example, Zurich's risk engineering teams conduct on-site audits for high-exposure infrastructure like power grids, using drone surveys and material testing to refine vulnerability curves and prioritise upgrades that could significantly reduce potential maximum losses.

Second, the insurance sector carries credibility with financial markets, driven by the industry's own experience and exposure to climate risks. Outputs from CAT models underpin insurance pricing, capital allocation decisions, reinsurance treaties and solvency regulation, and are familiar to investors, regulators and credit rating agencies. This can facilitate their adoption in resilience-linked financial products.

2.3.2 Quantifying avoided losses with insurance-grade risk assessment

To demonstrate the value of insurance sector risk assessment capabilities in quantifying avoided losses and unlocking resilience finance, we analysed annualised expected losses (AEL) and probable maximum losses (PML) with and without resilience measures for five power plants from diverse global regions: Africa (Senegal), Asia (Bangladesh), the Caribbean (Barbados), the Pacific (Indonesia) and Latin America (Brazil). These plants were selected to capture varying hazard exposures (such as floods, storms or heatwaves) and scales, offering an illustrative view of energy infrastructure risks. The names of the power plants have been anonymised, as the analysis is based on generalised parameters rather than detailed site-level engineering data. The purpose of this assessment is to provide an estimate of the scale of losses that can be avoided when resilience is embedded in infrastructure through probabilistic risk assessment, rather than to detail the exact vulnerability of individual facilities.

Our approach to assessing avoided losses combined physical damage, downtime losses and revenue disruption under very high-intensity hazards. The hazard probabilities for the sample power plants were derived using Zurich Resilience Solutions' proprietary Climate Spotlight tool, which provided the hazard probability for the exact location coordinates of the sample plants (see Box 2 for a summary of the methodology).

Although the five plants differ markedly in investment size and energy output, the analysis revealed substantial avoided losses from risk-reduction measures. Table 1 (page 20) shows AEL per power plant and project these losses over 10, 15 and 20 years (adjusted for inflation).

BOX 2. METHODOLOGY AND ASSUMPTIONS

To assess the value of resilience investments in energy infrastructure, we modelled expected losses with and without resilience measures for five representative power plants across Africa, Asia, the Caribbean, the Pacific and Latin America. The analysis combined physical damage and revenue disruption using the following formula,

$$L_{event} = (DR \times V_{infra} \times p) + (P_{MWh} \times G_{day} \times G_{down})$$

where losses reflect the product of hazard probability (*p*), infrastructure value (*V_infra*), damage ratio (*DR*), average electricity price (*P_MWh*), daily output (*G_day*) and downtime (*G_down*).

Key steps and assumptions included:

Hazards assessed: Sixteen types, including floods, windstorms, storm surge, heatwaves, wildfires and earthquakes. Probabilities were generated using Zurich Resilience Solutions' proprietary Climate Spotlight tool for 'very high' intensity events across 2020, 2030 and 2050 under Shared Socioeconomic Pathways 2–4.5 climate scenarios.

Distributions applied: The damage ratio followed a Beta distribution (mean 0.35, range 0.1–0.6, α=2, β=4). Downtime followed a triangular distribution (5–90 days, with a mode of 30). Monte Carlo simulations (10,000 iterations) captured the range of outcomes.

Resilience effect: Based on the 2019 World Bank Group study on strengthening new infrastructure assets by Hallegatte et al. and combining their modelling outcomes with our analysis, we found that cost-effective resilience measures were found to reduce both damage and downtime losses in power plants by more than 80%.

Outputs generated:

Annualised expected loss (AEL): the mean across simulations.

Probable maximum loss (PML): the 90th percentile.

Projections were made over 10, 15 and 20-year horizons, applying a 5% annual inflation adjustment.

Global scaling: The framework was extrapolated to the global power sector, which is valued at US$8.9 trillion. Daily generation (85.6 TWh) and an average wholesale price of US$100/MWh were applied, with hazard probability fixed at 0.01.

Full details of the methodology, parameters and data sources are provided in Appendix 1.

For all five plants, AEL without resilience averages to US$48.6 million annually, reducing to US$9.8 million with resilience measures, or US$38.8 million in savings per year. Over 20 years (adjusted for inflation), cumulative avoided losses reach US$10.34 billion (see Figure 11).

Tail risks also show substantial reductions. PMLs without risk-reduction measures reach as high as US$221 million annually. Resilience investments can reduce this to US$44 million, avoiding US$177 million in potential extreme event losses each year (see Figure 12). By reducing both expected and peak losses, resilience interventions demonstrate a triple dividend: steady fiscal savings, reduced business interruption and enhanced protection against catastrophic shocks.

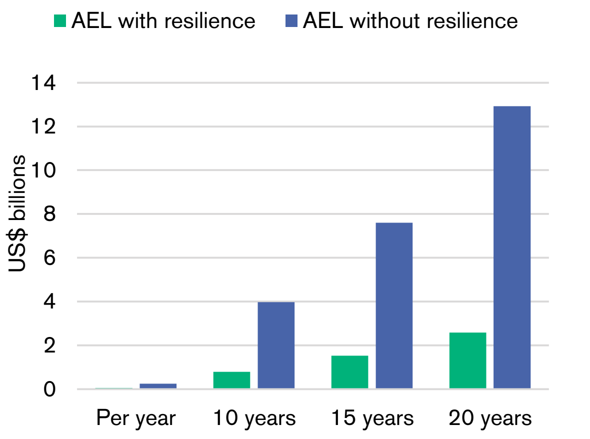
Globally, for the power sector's US$8.9 trillion replacement value, AEL without resilience is US$547.12 million annually, rising to US$29.09 billion over 20 years (see Figure 13). Resilience measures can reduce this to US$5.82 billion, thereby avoiding US$438 million yearly (or US$23.27 billion over 20 years). This illustrates the sector's vulnerability but also the transformative potential of resilience, where upfront investments based on risk assessment can yield returns of four to six times the investment through avoided damages and business continuity.

Table 1. Characteristics of the sample power plants and AEL with and without risk-reduction measures

Country location of sample power plant	Actual investment (US$ millions)	Daily energy output (MWh)	Price/ MWh (US$)	AEL figures (in US$ millions) with and without resilience							
				Annual		10 year*		15 year*		20 year*	
				Without resilience	With resilience	Without resilience	With resilience	Without resilience	With resilience	Without resilience	With resilience
Indonesia	800	6,760	80	23	5	372	74	713	143	1213	243
Bangladesh	12,650	51,840	90	204	41	3,317	663	6,351	1,270	10,807	2,161
Brazil	3,160	5,318	31	10	2	168	34	322	64	549	110
Barbados	175	153	208	1	0	24	5	47	9	79	16
Senegal	342	1	110	5	1	86	17	165	33	281	56
World	8,900,000	85.63 TWh	100	547	109	8,912	1,782	17,061	3,412	29,093	5,819

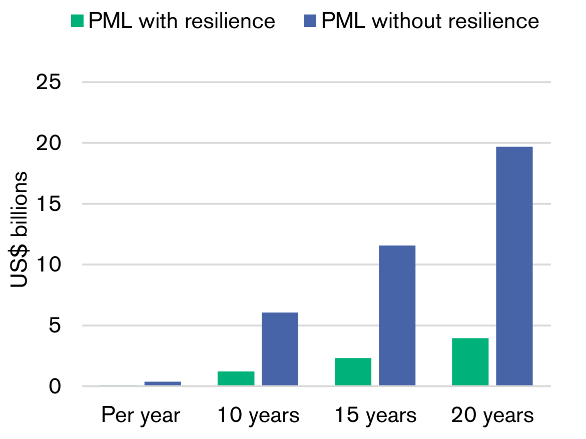
* adjusted for inflation.

Figure 11. AEL for sample power plants



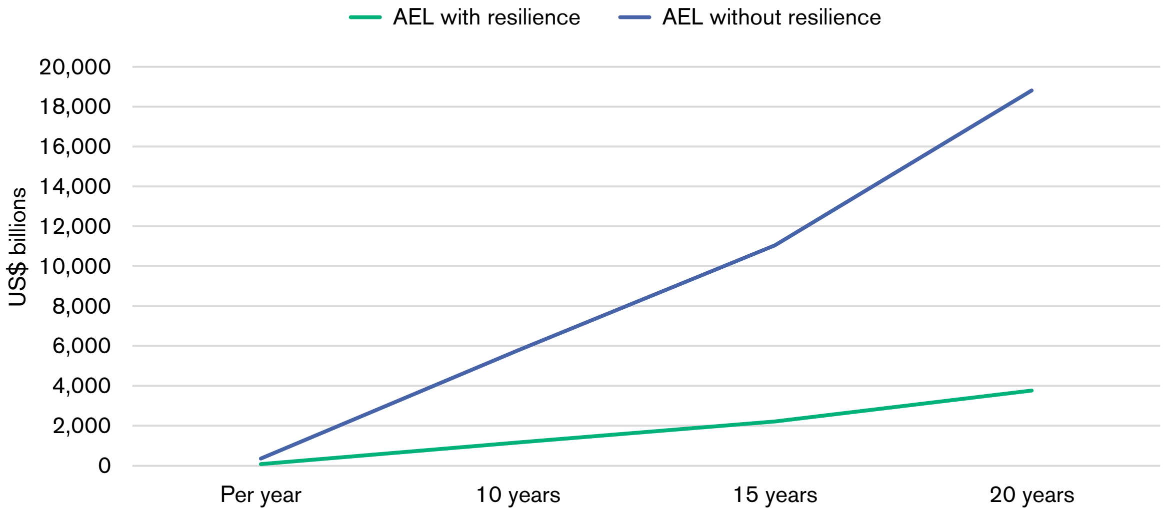
Source: Authors' calculations.

Figure 12. PML for sample power plants



Source: Authors' calculations.

Figure 13. Global AEL for the power sector (with and without resilience)



Source: Authors' calculations.

Looking at the broader infrastructure pipeline spanning sanitation, electricity, transport and telecommunications, the G20 Global Infrastructure Outlook projects that US$94 trillion is needed by 2040, with US$18 trillion unfunded.[71] Assuming similar 80% loss reductions, resilience measures could avoid US$7.5–9.6 trillion in cumulative losses, effectively unlocking the unfunded portion by significantly de-risking projects, thereby attracting private and MDB capital. In electricity (including power), savings alone could cover 20% of the gap, while transport infrastructure and water/sanitation could add another 30%. This highlights the scale of the financial benefits achievable through sector-wide adoption of risk assessments to future-proof investments.

These findings echo experience from real-world case studies. For example, Box 3 details Zurich Resilience Solution's work with AUDI AG on flood resilience.

The AUDI case underscores how advanced risk analytics and engineering assessments can transform resilience from a perceived compliance cost to a driver of business continuity. The high financial return is also demonstrated by a counterfactual context. When Chennai, one of the world's largest automotive manufacturing and supply chain hubs, was hit by flooding in 2015, the lack of adequate resilience measures caused widespread business interruption. Multiple global companies, including Hyundai, Ford and BMW, were forced to suspend production for several days as plant facilities, warehouses and the essential component supply belt were inundated by water. The economic fallout of this was disastrous. While overall economic losses in the region exceeded US$3 billion, the industry sector alone was estimated to have suffered a production loss of up to Rs 1,500

BOX 3. CASE STUDY: FLOOD RESILIENCE ASSESSMENT FOR AUDI AG (VOLKSWAGEN GROUP), GERMANY

In May 2016, torrential rains in southern Germany caused severe flooding that disrupted production at AUDI's Neckarsulm plant. In response, AUDI partnered with Zurich Resilience Solutions to better understand its vulnerabilities and strengthen future preparedness. Zurich applied probabilistic catastrophe modelling, on-site engineering inspections and climate scenario analysis to identify major risks, including water ingress, supply chain disruptions and embankment erosion. Based on these findings, AUDI, in collaboration with local authorities, implemented a series of measures such as inflatable water barriers, reinforced embankments, retention ponds, improved drainage systems and coordinated flood response exercises.

When another episode of heavy rainfall hit the region in June 2021, these interventions proved highly effective. Production continued with only minor disruptions, such as sewage overflow, avoiding the large-scale shutdown that occurred in 2016. By avoiding downtime, AUDI saved millions of dollars in potential losses, demonstrating that targeted resilience investments deliver a strong financial return.

Source: Zurich Insurance Group (2021) Staying afloat during floods: helping AUDI AG (Volkswagen Group), 29 November.

crore (US$225 million) due to downtime.[72,73] Unlike the AUDI plant, which avoided significant downtime through resilience measures, these manufacturers faced weeks of cleanup, supply chain delays and significant market disruption. This example demonstrates that the costs of delayed adaptation far outweigh the upfront investment required for risk-informed resilience.

This risk-informed approach is equally crucial for managing systemic climate threats in the urban environment, as demonstrated by the case study of extreme heat adaptation (see Box 4) in Madrid, Spain.

Madrid's strategy is expected to safeguard public health and economic continuity. Conversely, the tragic 2003 European heatwave overwhelmed the city of Paris, resulting in an estimated 735 heat-related deaths in the capital alone.[74] Across France, this systemic failure led to an estimated 15,000 heat-related deaths and caused an estimated €13.2 billion (US$16 billion) in total economic loss due to overwhelmed health systems, infrastructure breakdowns and labour disruption.[75,76] The catastrophic, unbudgeted cost of this failure to proactively invest dwarfs the manageable, strategic investment made by cities like Madrid to enhance their long-term resilience.

The analysis of both the industrial (AUDI/Chennai) and urban (Madrid/Paris) case studies demonstrates the potential value that can be achieved through comprehensive risk assessment. Tools like the modelling software that Zurich has developed, which combine hazard probabilities, loss data and risk engineering insights, could provide a more robust framework to guide investment decisions on specific resilience actions. This type of modelling can be used to inform quantification of the economic benefits and savings achievable through specific resilience measures, ultimately allowing policymakers and financiers to move adaptation spending out of the emergency budget and into the high-return investment portfolio.

BOX 4. CASE STUDY: EXTREME HEAT ADAPTATION IN MADRID, SPAIN

Madrid faces mounting risks from extreme heat, with urban temperatures sometimes exceeding those of surrounding rural areas by up to 8.5°C due to the urban heat island effect. Recognising the growing threat to public health, infrastructure and economic activity, Madrid City Council collaborated with Zurich Resilience Solutions to develop a resilience strategy. Zurich combined climate projections with vulnerability and cascading impact analyses, examining how heat waves would affect health systems, water and energy supplies and transport networks, with particular attention to vulnerable groups such as children and older people.

The resulting roadmap, integrated into the city's 'Madrid 360' strategy, prioritised a mix of nature-based and risk-reduction measures. These included expanding green spaces and parks, planting more trees, creating cooling water features, introducing reflective building materials and constructing shaded sidewalks, alongside adaptive approaches, such as adjusted work hours, resident education and preparedness planning for extreme heat events. By addressing risks systemically, the plan reduces health hazards, service disruptions and infrastructure stress, while also enhancing fiscal stability. Framed as both a public health intervention and an economic resilience strategy, the assessment positioned cooling measures as bankable investments with high cost–benefit ratios.

Source: McAllister, S (2025) Madrid's battle against extreme heat: how Zurich is strengthening the city's climate resilience, Zurich Insurance Group, 21 July.

How to embed risk analytics by creating a resilience finance ecosystem

The evidence in the previous section highlights the value of better risk data and how the insurance sector's existing capabilities can deliver risk insights that can help drive investments into resilience. This section examines how governments, MDBs and the private sector can harness these capabilities to inform their decisions and support greater resilience finance. We outline the challenges that need to be addressed, particularly in emerging markets and developing economies (EMDEs), and present the roles different actors can play to build an enabling resilience finance ecosystem.

3

3.1 How insurance sector risk analytics can be used by different actors

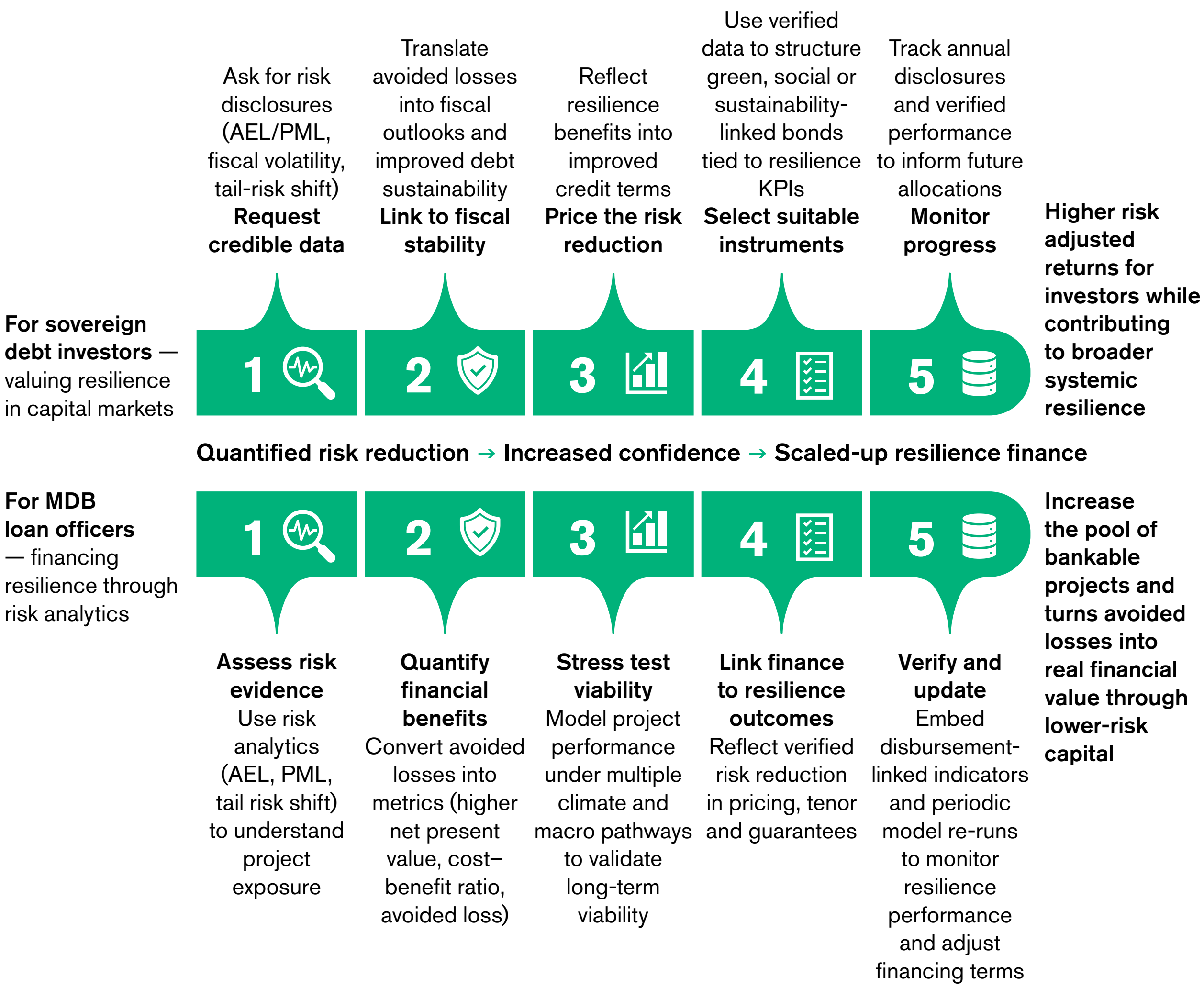
Investing in resilience is essential, but amid competing priorities for fiscal resources and capital, its value needs to be clearly established. Quantifying risks and the impacts of risk-reduction measures in terms of avoided losses can provide the necessary evidence base for this, helping decision makers identify the most efficient and optimum options for reducing losses. Governments, MDBs and private sector actors can achieve this by using insurance risk assessment as follows:

- Many governments attempt to raise resilience finance through green, social or sustainability bonds, but they often struggle to provide standardised, verifiable metrics demonstrating the financial return of resilience projects. By embedding insurance risk insights into sovereign bonds, governments can present quantifiable evidence of estimated avoided losses and higher returns. For instance, when the District of Columbia Water and Sewer Authority (DC Water) issued a US$25 million Environmental Impact Bond in 2016,[77] they used probabilistic modelling of stormwater runoff risks to fund green infrastructure projects. The investor payouts were adjusted based on performance metrics, such as reduced sewer overflows and avoided environmental damage, monetising avoided losses through higher or lower returns tied to verified outcomes.[78] This reassured investors and provided the issuer with upfront capital for resilience without increasing debt burdens, while indirectly enhancing creditworthiness for broader sovereign or municipal financing.
- MDBs and development banks are under pressure to scale up adaptation finance but are often cautious about investing in infrastructure in climate-exposed settings, due to concerns over returns and long-term viability.[79] Probabilistic catastrophe modelling can allow them to stress-test projects against future extremes and demonstrate their long-term viability. For instance, the World Bank's Disaster Risk Financing and Insurance (DRFI) programme has used catastrophe modelling to unlock more than US$5 billion in parametric coverage for more than 50 countries, cutting reliance on post-disaster borrowing by 20–30%.[80] Similarly, the Asian Development Bank's US$1 billion resilient infrastructure programme in Asia mandated site-level vulnerability assessments, which expanded the viable project pipeline by 30% and reduced projected losses in flood-prone regions.[81] These approaches can give MDBs confidence to de-risk projects, increase the likelihood of affordable insurance cover and structure blended finance investments that can attract private co-financing. European institutions have also begun adopting these tools. The European Investment Bank (EIB), for example, employs Munich Re's Climate Financial Impact models to forecast asset-level losses and integrate them into blended finance structures. This approach demonstrates to private investors that risk-reduction measures can generate returns of between two and five times the investment, thereby attracting co-financing.[82,83] By requiring catastrophe-based risk assessments in loan approvals and integrating them into economic analysis, MDBs can treat resilience as a core element of the lending process rather than an optional safeguard.
- The private sector is increasingly recognising climate risks in its investment decisions, but data is not always available at the right level of granularity to inform resilience investments on a standalone basis. Insurance analytics can fill that gap by showing, for example, how measures such as flood defences might reduce business interruption, reducing production downtime, protecting revenues and ultimately improving ROI. Asset managers and institutional investors are also beginning to integrate these climate risk assessment tools. However, they could extend their use to make the case for investing early in risk reduction. For example, if investors had such quantified risk-reduction metrics accompanied by cost–benefit analysis, they could demand resilience safeguards in the infrastructure investments they finance, thereby protecting their longer-term portfolio value, as well as revenues. Analyses by Zurich Resilience Solutions of institutional investment portfolios have shown that embedding resilience into investment decision making can reduce exposure to climate risk while supporting industry-wide sustainable practices.[84]

These applications show how embedding insurance risk insights in decision making can help break the cycle of underinvestment and support the scale of capital needed for resilient development. To demonstrate how this might work in practice, we have illustrated two scenarios showing how MDBs and sovereign debt investors could translate quantified risk reduction into financial value (see Figure 14) — one through loan structuring and concessional terms, the other through improved credit pricing and capital market access. While their entry points would differ, both pathways could converge on the same outcome: transforming risk transparency into increased investor confidence, which could then translate into scaled-up resilience finance.

This step-by-step framework shows that when resilience performance is measured, verified and priced, it can become investable, providing a pathway for scaled-up and sustained flows of climate resilience finance.

Figure 14. MDB and investor pathways to using risk analytics to unlock resilience finance at scale



3.2 What issues will need to be addressed to make this work?

To realise the potential of insurance sector risk assessment capabilities in scaling resilience finance, several critical issues must be addressed to facilitate effective implementation and adoption. These issues span technical, institutional and financial dimensions, particularly in vulnerable economies where data scarcity and capacity constraints are most acute.

A primary concern is the persistent lack of high-resolution hazard and asset data in many developing countries, which limits the accuracy of risk assessments. The International Development Finance Club (IDFC) has launched a three-year capacity-building programme, facilitated by its Climate Finance Facility, to enhance data collection and green finance tracking among member institutions, targeting US$1.3 trillion in cumulative commitments by 2025.[85] Nonetheless, at present, many developing and low-income countries still lack comprehensive, high-resolution asset inventories, severely hampering site-level vulnerability scoring essential for effective risk assessments. This data gap is being addressed through partnerships with the insurance sector to develop shared data platforms and open-source tools. A key example is the Open Data for Resilience Initiative (OpenDRI) operated by the Global Facility for Disaster Reduction and Recovery (GFDRR), which has actively improved data access in more than 35 countries since its launch in 2011, using technologies such as satellite imagery and crowdsourcing to map assets.[86] The Insurance Development Forum (IDF) is also working towards improved risk understanding, including through the launch of free open-source risk modelling tools in April 2025, supporting access to climate and disaster risk analysis.[87] Equitable access to these tools, however, remains uneven, risking the exclusion of the most vulnerable countries. The IDF's capacity-building efforts, as detailed in its 2025 report, aim to address this by partnering with local institutions to co-design open access tools, with early progress demonstrated in targeted EMDEs such as Ghana, Bangladesh and the Philippines.[88]

Another issue is limited technical capacity to interpret and apply complex risk analytics, particularly in emerging markets and LDCs. This capacity gap delays adaptation progress, as noted in the 2024 UNEP Adaptation Gap Report, and is compounded by the significant resources required for large-scale training across the Global South.[89] The IDF has embedded capacity building into its risk modelling work, which includes training programmes that help public sector officials to improve their risk understanding and learn how to use risk models to enhance their decision-making processes. However, scaling up these training programmes remains a challenge without sustained and dedicated funding.[90] Addressing this requires philanthropies and climate funds to cover the cost of deploying risk assessment tools, including continuous capacity building and hands-on support, allowing their value to be demonstrated in real market settings. For example, partnerships such as the African Development Bank's collaborations with insurers and reinsurers to implement advanced flood risk models have already shown how such tools can reduce project appraisal times and approval.[91]

Financial constraints remain a major barrier, particularly the high upfront costs of adopting advanced risk assessment tools. In many EMDEs, where public budgets are already strained, investing in sophisticated risk analytics can appear prohibitive. Some recent initiatives are beginning to demonstrate how these challenges can be addressed. For instance, innovative programmes like the Global Risk Modelling Alliance,[92] a public–private partnership that funds country access to risk modelling tools, and the World Bank's Southeast Asia Disaster Risk Insurance Facility (SEADRIF) have successfully reduced costs and maintained operational resilience by integrating analytical capabilities provided by private insurers, which facilitates private sector co-financing.[93]

Institutional resistance and regulatory misalignment can also complicate adoption. A 2023 PwC Global Risk Survey highlights the fragmentation and institutional barriers public sector entities face in managing risk and integrating external data, creating silos that delay policy alignment.[94] For MDBs, this institutional challenge often manifests as a lack of in-house expertise in applying climate risk assessment results. This, combined with limited high-quality data in many countries, hinders integration into project appraisals. The insurance sector is well-positioned to address this through awareness-building efforts, such as workshops and partnerships that demonstrate the value of probabilistic models, while providing enhanced datasets on risks to inform MDB decision making. The IDF advocates for regulatory sandboxes (controlled environments for piloting innovative financial solutions), which have been tested in locations like Kenya and Bangladesh to pilot insurance risk analytics-integrated frameworks that streamline and accelerate approval processes.[95] Further, intellectual property concerns around proprietary models, as noted by the IDF, can limit data-sharing unless insurers are incentivised through public–private partnerships.[96]

While different efforts are ongoing, addressing these issues holistically requires a multi-pronged strategy, which should include: enhancing data infrastructure through global partnerships, building capacity via targeted training programmes, offsetting costs with innovative financing, aligning regulations through pilot frameworks and ensuring equitable deployment to maximise societal benefits. Only by tackling the obstacles can the ecosystem fully harness insurance sector risk assessments and risk management insights to transform resilience into a global priority.

3.3 Establishing a thriving ecosystem for resilience finance

To improve the use of risk analytics and enhance resilience finance, a thriving ecosystem requires the coordinated and complementary integration of clear roles, supportive rules (policy and regulations), effective tools and necessary capital. Governments can provide the foundational rules through enabling policies. MDBs and development banks can contribute both rules and capital via financing mechanisms. Insurers can supply analytical tools like modelling platforms, as well as risk reduction and risk engineering advisory services, while the private sector can mobilise capital and innovation for implementation.

Such an ecosystem can position risk assessment as a public good, enabling actors to scale resilience from isolated projects to systemic transformation, aligning incentives to close the resilience financing gap. The IDF and Bridgetown Initiative also call for the integration of risk analytics into national planning and MDB decision making to enhance resilience finance and support vulnerable economies.[97] But for it to develop, stakeholders must contribute meaningfully, focusing on improving data development and access. Lessons from other sectors show that such coordination can be successful when different stakeholders come together and reinforce each other's strengths.

Governments must establish the policy foundation and drive implementation to ensure that resilience becomes a national priority. Their actions could include:

- Mandating comprehensive risk modelling and integration of risk-reduction measures in project appraisals for all public investments, ensuring forward-looking assessments of tail risks to safeguard critical infrastructure.

- Embedding risk analytics-based resilience metrics, such as loss exceedance curves, in sovereign bond frameworks to signal lower volatility and attract better financing terms.
- Using modelling outputs to strengthen prospectuses for resilience, green, social and sustainability bonds, quantifying benefits, such as returns on ecosystem-based defences, to appeal to global investors.
- Integrating risk analytics into strategic national planning for national adaptation plans, nationally determined contributions and municipal plans, to prioritise resilience investments based on quantified risks and avoided losses.

Lessons from the renewable energy sector show how governments can create enabling environments to drive investment. Policies like competitive auctions and public finance de-risking tools have reduced solar costs (specifically, the global weighted average levelised cost of electricity (or LCOE) by 90% since 2010 and attracted more than US$2.6 trillion in global investments through 2023.[98] India's renewable energy sector exemplifies this: support mechanisms, such as the Production-Linked Incentive scheme and tax holidays, grew installed renewable capacity from about 18 GW in 2010 to more than 190 GW by September 2025,[99] drawing private players like Adani and banks like the State Bank of India to collaborate on green loans, lowering prices and boosting foreign direct investment. Similar policy and regulatory incentives can help scale up the use of risk analytics to increase resilience investment.

MDBs and development banks are essential in providing financing and technical expertise by bridging the gap between risk assessment and actionable projects. Their contributions could involve:

- Requiring risk modelling and integration of risk-reduction measures for infrastructure loans (including the potential introduction of resilience-linked rates) to verify long-term viability, ensuring projects withstand future extremes.
- Leveraging catastrophe-based analyses to expand the pipeline of bankable resilience projects, prioritising those with better cost–benefit ratios to maximise impact.
- Supporting countries in developing open access risk models through partnerships with insurers, facilitating public planning and decision making.
- Integrating insurance mechanisms in their risk layering and financing strategies.

The Green Economy Financing Facility (GEFF) operated by the European Bank for Reconstruction and Development (EBRD) offers valuable learning for a comparable approach. By mandating energy efficiency audits and providing incentives like grants and top-ups, the GEFF has unlocked more than €6.3 billion in EBRD green financing, reaching more than 231,000 clients (small businesses and homeowners) and achieving annual CO_2 emissions reductions exceeding 10 million tonnes.[100] Integrating risk assessments and risk-reduction measures into such financing mechanisms could de-risk infrastructure projects and drive resilience investment.

Insurers can provide the analytical and advisory foundation, offering expertise and tools to scale investment in resilience beyond traditional boundaries. Their roles could include:

- Offering technical assistance to apply risk assessment tools beyond traditional contexts, such as providing training to public planners on data fusion techniques to enhance local capacity and integrating future climate risk assessments into building codes and planning zones.
- Partnering with governments and investors to co-create resilience performance benchmarks, using standardised vulnerability scoring for cross-sector comparisons to build trust.
- Developing shared platforms for real-time hazard data, addressing gaps in high-resolution inputs.
- Scaling up risk engineering capabilities through capacity-building initiatives, providing risk reduction and risk advisory services and knowledge-sharing to reduce risks and enhance climate resilience in vulnerable regions and planned infrastructure investments.

The private sector can drive innovation and implementation by leveraging risk insights to protect assets and supply chains. Their actions could include:

- Incentivising resilience assessments in supply chains via internal mandates or supplier requirements, using scenario simulations to evaluate disruptions and prioritise upgrades.
- Reporting risk analytics-based resilience metrics, such as projected downtime reductions, in ESG disclosures to attract sustainability and impact-focused capital, enhancing market competitiveness.
- Linking supplier financing rates to demonstrated improvements that implement findings from risk modelling.

Examples from the private sector, such as IKEA's long-term commitment to sustainable cotton, show how this can work in practice. The company co-founded and expanded the Better Cotton Initiative (BCI), which has trained hundreds of thousands of farmers in sustainable methods, enabling IKEA to achieve 100% sustainable cotton sourcing since 2015.[101] This shift resulted in more efficient farming practices, including significant water and chemical use reductions, demonstrating the power of private sector mandates to transform supply chains. Overall, the BCI standard has contributed to the sustainable management of 2.5 million hectares of farmland, which represents 22% of global cotton production.[102] Similar mandates for risk analytics could be integrated by private actors to mobilise resources and incentivise resilient supply chains.

To summarise the complementary nature of these roles, Table 2 outlines how they interlink to form a cohesive ecosystem.

These roles, when pursued collaboratively, can overcome persistent challenges like data silos and limited technical capacity by creating verifiable pathways for risk transfer and value creation. For example, the experience of Caribbean nations demonstrates how the integration of insurer-provided catastrophe modelling into national strategies informed significant targeted investments in resilience and established benchmarks that MDBs continue to reference for regional lending.[103] Private actors could similarly integrate risk analytics to mobilise resources and incentivise resilient supply chains.

Table 2. The building blocks of a cohesive resilience finance ecosystem: roles and contributions of key actors

ACTOR	ROLE	RULES (POLICY/ REGULATION)	TOOLS	CAPITAL
Governments	Set enabling environment; integrate resilience in planning and budgets	Mandate probabilistic risk modelling in public investment appraisals; embed resilience metrics in sovereign/municipal bond frameworks; require disclosure of avoided loss and stability gains	Vulnerability scoring platforms; national risk observatories; open access hazard and exposure data	Sovereign/ green/ social/sustainable/ resilience bonds; budget allocations
MDBs and development banks	Finance, de-risk and scale resilience projects	Require catastrophe-based modelling for infrastructure loans; standardise resilience metrics in economic analysis; support open access models for borrowers	Probabilistic catastrophe models; stress testing frameworks; project preparation, technical assistance	Concessional loans, guarantees for blended finance structures
Insurance sector	Supply expertise, models and verification	Develop and co-own resilience performance benchmarks; align with ESG disclosure standards for standardised reporting of climate-related financial impacts, enabling integration of risk analytics into disclosures	Probabilistic catastrophe models, business interruption/ cascading impact modules, site-level risk engineering, data fusion platforms	Risk transfer products (parametric covers, resilience bonds); premium credits for verified resilience
Private sector (corporates, investors, lenders)	Implement and invest in resilience; safeguard continuity	Integrate risk analytics-based resilience metrics in ESG reporting and loan agreements; supplier financing tied to resilience performance	Scenario simulations; supply chain business interruption analysis; asset-level risk audits	Corporate capex; infrastructure equity; supplier financing; resilience-linked loans/bonds

Looking forward

The Baku to Belém Roadmap to 1.3T launched at COP29 aims to mobilise US$1.3 trillion in annual adaptation and resilience finance by 2035. Achieving this requires a fundamental shift in the global financial system, including reforming multilateral systems and unlocking private capital. A key step is for policymakers and investors to view resilience as a mainstream, investable asset class. This section summarises three ways in which governments, MDBs and private finance can leverage the insurance sector's tools and expertise to create a thriving resilience finance ecosystem that can deliver climate resilience at scale.

4

The ambitious commitment to mobilise US$1.3 trillion in annual adaptation and resilience finance by 2035[104] requires a fundamental shift in global finance. The Baku to Belém Roadmap to 1.3T, launched under the COP presidencies of Azerbaijan and Brazil, is the political pathway for achieving this scale. The process has an explicit focus on reforming multilateral systems and unlocking private capital, providing a structured framework from COP29 through to COP30 and beyond to identify and implement mechanisms for scaling finance.

What matters most in the immediate term and for the continued success of the Roadmap is moving from pilot resilience investment to a system where resilience is treated as a mainstream, finance-ready asset class. That means embedding risk analytics into the rules that shape markets. In other words, how sovereign bonds are structured, how MDBs appraise projects and how private investors evaluate portfolios. The opportunity lies in showing that resilience is not only a defensive measure but a credit-enhancing, volatility-reducing and value-creating proposition that belongs at the centre of financial decision making.

Three priorities stand out for realising this opportunity under the Baku to Belém Roadmap. First, governments need to use insurance risk insights to build credibility with markets and rating agencies. By quantifying avoided losses and stability gains, countries can issue resilience, social, sustainability or green bonds on stronger terms, as well as bring more bankable proposals to the Green Climate Fund, the FRLD and MDB pipelines.

Second, MDBs must standardise the use of probabilistic modelling and risk-reduction planning in project preparation and lending, so that resilience becomes a built-in feature of their operations rather than an optional safeguard. This shift would facilitate blended finance by enabling guarantees to be precisely sized (for example, against the 'tail' risks), thereby de-risking projects and attracting more private capital.

Third, the private sector should integrate climate risk assessments and risk-reduction metrics into ESG frameworks and investment due diligence, thereby rewarding firms and projects that can demonstrate quantifiable risk reduction and business continuity benefits.

For this ecosystem to mature, insurance sector analytics should be made more accessible through collaborative partnerships. Making non-sensitive hazard and exposure data widely available and developing clear verification protocols for that data are essential first steps. To ensure countries can then apply these tools consistently and at scale, regional hubs and training partnerships should provide technical support for first-mile data collection (gathering initial high-resolution hazard and asset data). Equally, climate funds and MDBs can help offset the upfront costs of risk modelling and risk management so that they are not a barrier to entry.

By embedding these approaches across sovereign finance, MDB operations and private investment, resilience can move from being underfunded and undervalued to being investable at scale. The Baku to Belém Roadmap to 1.3T has the potential to support such action and fundamentally reshape global financial architecture — by embedding climate risk assessment into the core operational procedures of the financial system.

Appendix 1. Methodology for loss and resilience benefit estimation

We estimated the benefits of resilience investments by comparing expected losses under baseline and resilience-enhanced scenarios for power generation plants exposed to climate and natural hazards. Total losses from a disaster event were modelled as the sum of infrastructure damage[105] and lost revenue due to downtime:[106]

$$L_{event} = (DR \times V_{infra} \times p) + (P_{MWh} \times G_{day} \times G_{down})$$

where *DR* is the damage ratio, *V_infra* the infrastructure value, *p* the hazard probability, *P_MWh* the average wholesale electricity price, *G_day* daily energy output and *G_down* the downtime in days.

Plant-specific infrastructure values and generation capacities were obtained from project documents, while electricity prices reflected prevailing wholesale rates. Hazard probabilities were derived from a dataset covering 16 hazards (including flood, windstorm, storm surge, heatwave, wildfire and earthquake) across three time horizons (2020, 2030 and 2050) and four Shared Socioeconomic Pathway climate scenarios. Only "Very High" intensity events were retained for the loss analysis. Loss modelling used a Monte Carlo simulation (a computational technique that uses repeated random sampling to obtain numerical results) with 10,000 iterations. Damage ratios[107] were sampled from a Beta distribution (mean 0.35, range 0.1–0.6, $\alpha=2$, $\beta=4$) and downtime from a triangular distribution (min. 5 days, mode 30 days and max. 90 days). Although we did not find an exact published distribution with those parameters, empirical and survey studies report that infrastructure and business operations are often disrupted for weeks to months after a disaster. (For example, a study by Watson et al. in 2024 found business interruption durations in survey data that lasted multiple weeks.[108]) For each iteration, infrastructure loss was computed as *DR* × *V_infra* × *p* and revenue loss as *P_MWh* × *G_day* × *G_down*, where *p* is the hazard probability. The sum of these components yielded the total loss distribution.

To quantify the benefits of resilience, avoided losses were estimated by assuming that resilience investments reduce losses by 80%, consistent with prior empirical evidence from an infrastructure risk-reduction study published by the World Bank Group.[109,110] Thus, losses without resilience and with resilience were compared to estimate the benefit of resilience investment.

From this, the annual expected loss (AEL) was taken as the mean and the probable maximum loss (PML) as the 90th percentile. These were projected for 10, 15 and 20-year horizons using a 5% annual inflation adjustment. The benefit of resilience investment was measured as the avoided AEL and PML between non-resilient and resilient cases.

For the global estimation of resilience benefits, the same loss modelling framework was extended from the plant-level to the entire power generation sector. Infrastructure replacement value was derived by multiplying the installed global generation capacity[111] by the average capital cost per unit of capacity,[112] representing the replacement value of the worldwide fleet of power plants. Daily electricity generation was based on total global electricity output for 2024,[113] converted to an average per day figure, while the average wholesale electricity price[114] was applied uniformly across all plants. Damage ratios, downtime distributions and hazard probabilities were retained from the plant-level analysis to ensure consistency, with hazard probability fixed at 0.01 for very high-intensity events. Using these inputs, global AEL and PML were calculated following the same Monte Carlo simulation procedure.

This analysis is illustrative and based on assumptions. It does not rely on site-specific engineering or financial data, but on assumptions and representative global datasets. The results are intended to demonstrate the potential order of magnitude of resilience benefits, not to prescribe precise vulnerability levels for specific plants or facilities. Findings should therefore not be applied directly to investment or operational decisions at the site level.

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Bharadwaj, R, Karthikeyan, N, Kumar, B and Mitchell, T (2024) Redefining credit ratings for Small Island Developing States: a pathway to climate resilience and economic stability. IIED, London. www.iied.org/22471iied

Climate change is intensifying economic losses worldwide, with the greatest impacts in countries least able to manage them. Yet despite evidence that resilience investments deliver economic and fiscal returns, finance for resilience remains far below what is needed. This paper shows how resilience can be transformed from a perceived cost into an investable asset. The insurance sector's risk analytics and engineering expertise can quantify avoided losses, identify resilience-building opportunities and demonstrate financial returns. Integrating these insights into sovereign finance, multilateral development bank operations and private investment can turn resilience from an undervalued priority into a mainstream, finance-ready asset that is investable at scale.

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